Synthesis and Detection Algorithms for Oblique Stripe Noise of Space-Borne Remote Sensing Images

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***Abstract*— Oblique stripe noise widely appears in remote sensing images after image correction, exhibiting arbitrary tilt angles and parallel distribution. Because of its arbitrary randomness in tilt angles and lengths, oblique stripe noise increases the difficulty of detection compared to vertical or horizontal stripe noise. For the first time, we propose a group of oblique stripe noise synthesis and detection algorithms combining imaging mechanisms and deep learning. To get controllable synthetic oblique stripe noise data for the training detection model, two sample augmentation methods are presented by the image correction's imaging mechanisms with new linear transformation and the generative adversarial network (GAN) algorithm with CycleGAN, respectively. A large-scale simulated stripe noise dataset simulated oblique stripe noise dataset (SOSD) is simulated using these two methods. A new deep learning detection algorithm, robust detection of oblique stripe noise (RDOS), is presented considering the presence of oblique stripe noise. RDOS is trained using both SOSD and a real stripe noise dataset, and it obtains the optimal detection model for testing. The experimental results show that the accuracy reaches 82.93%, the recall rate reaches 85.17%, the $F1$ score reaches 84.04%, the average precision (AP) reaches 82.34%, and the frames per second (FPS) reaches 33.33. Compared with the general line detection models, our model exceeds ∼300% in accuracy and ∼60% in speed. In the future, the proposed algorithms will have great potential for application in various areas, such as quality evaluation, image preprocessing, and engineering problems related to multiangle linear object augmentation and detection.**

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I. Introduction

REMOTE sensing images play a crucial role in various applications such as land use classification, environmental monitoring, and urban planning [1], [2]; however, these images are often affected by various sources of noise and distortions, significantly reducing their quality and hindering accurate interpretation and analysis. One common type of noise that appears in remote sensing images is called "stripe noise" [3], [4]. Stripe noise refers to the presence of striped patterns in the images, which exhibit arbitrary tilt angles and parallel distribution after image correction, as shown in Fig. 1. The presence of stripe noise within remote sensing imagery significantly degrades data quality, affecting data interpretation and usability [5]; therefore, accurately detecting stripe noise is crucial for promptly identifying quality issues and aiding in the identification of problematic elements. Additionally, it also facilitates warnings to the preprocessing system for relative radiometric recalibration, holding significant academic and practical value.

In recent years, the detection of stripe noise has gained attention from researchers, and various methods have been proposed to address this problem by combining knowledge of image correction mechanisms and advancements in machine learning techniques [6], [7], [8]. From a technical perspective, these methods can be broadly categorized into traditional methods based on imaging mechanisms and deep learning methods.

Traditional methods require manual feature definition and often suffer from uncertainties in threshold settings [9], [10], [11], [12], [13]. For instance, Sun et al. [14] used the $\ell_w 2,1$-norm to characterize the joint sparsity (structural property) of stripe noise and used the iterative support detection (ISD) [15] to calculate the weight vector in the $\ell_w 2,1$-norm, thus facilitating stripe noise detection. Tsai and Chen [16] used the Canny algorithm to extract edge information from the images and combined the extract results with line-tracing algorithms to achieve stripe noise detection. Most of these methods are, however, applicable only to vertical/horizontal stripe noise detection issues and are not suitable for oblique stripe noise with a diversity of angle and length. Additionally,

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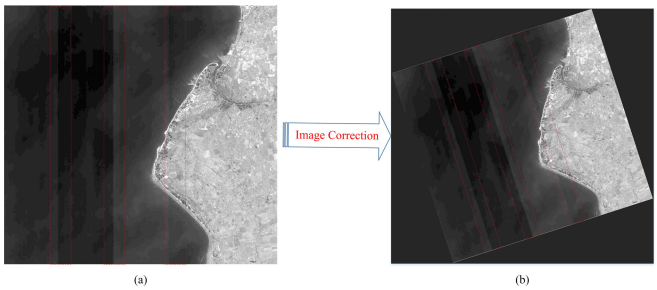


Fig. 1. Schematic of oblique stripe noise. (a) Stripe noise image without image correction. (b) Stripe noise image after image correction.

manual feature definition approaches limit their robustness and portability in stripe detection problems involving different sensor sources. On the other hand, deep learning methods consider stripe noise as a typical linear object, extracting high-dimensional meaningful semantics from data properties instead of relying on manual description and achieving good detection performance [17]. Deep learning, however, relies heavily on the quantity and distribution of data, and the manual annotation of a large amount of data incurs significant costs, making it challenging to apply in practical engineering applications [18], [19]. Oblique stripe noise detection, moreover, requires accurate prediction of stripe length and tilt angle, which requires further optimization in the design of deep learning detection algorithms compared to those designed for vertical or horizontal stripe noise.

To address the aforementioned issues, this study delves into two main aspects: data augmentation and object detection. We aim to construct a group of oblique stripe noise synthesis and detection algorithms combining imaging mechanisms and deep learning and then acquire an oblique stripe detection model. First, by acquiring more oblique stripe noise data and constructing a dataset, our aim is to enable the detection model construction process to capture a greater amount of common information about stripe noise. Second, optimization efforts are required to enhance deep learning algorithms for the precise detection of oblique stripe noise, considering the challenges associated with data dependence and accurate prediction of stripe characteristics. Our main contributions are summarized as follows.

1) We propose a new linear transformation simulation method for simulating oblique stripe noise based on the imaging mechanism of its generation [20]. A large dataset of remote sensing images with simulated oblique stripe noise, referred to as the first part of the simulated oblique stripe noise dataset (SOSD), is generated using this method.
2) We employ the Cycle-GAN algorithm [21] to train and evaluate to build an oblique stripe noise generation model using gradient images obtained from blank images based on the above method and the first part of SOSD. The generation model is then used to simulate a new dataset of remote sensing images with oblique stripe noise, referred to as the second part of SOSD.
3) We propose a novel linear stripe noise detection (LSND) algorithm called "robust detection of oblique stripe noise" (RDOS). We combine a real stripe noise dataset with SOSD from the above two parts to create an experimental dataset. Based on the dataset, we train and evaluate to get an optimal parameter detection model using the RDOS algorithm. The experimental results demonstrate that the RDOS-based model achieves good performance across different detection metrics and exhibits excellent speed.

Compared to previous stripe noise detection methods, our methods not only address the issue of insufficient training data, enabling more accurate detection of oblique stripe noise, but also have the potential to be applied to diverse and heterogeneous data sources. The methods and highlights derived from this research, moreover, have significant implications in high-resolution remote sensing image quality assessment [22], image preprocessing techniques [23], and multiangle linear feature detection [24], [25], [26], among other engineering challenges. The rest of this article is structured as follows. Section II represents the current research status of stripe noise simulation and detection. Section III describes the proposed methods. Section IV presents the experimental results and analysis. Finally, Section V concludes this article and also discusses the impact and potential future directions of applying these techniques to other fields and applications.

II. Related Work

A. Stripe Noise Simulation

Stripe noise simulation is one of the crucial steps in the research of stripe noise detection and removal. Its research significance lies in providing real and controllable synthetic data for algorithm evaluation, model training, and parameter optimization. By simulating stripe noise in different scenarios, researchers can gain a better understanding of the characteristics of stripe noise, design more effective detection or denoising algorithms, and enhance the quality and usability of remote sensing images. Currently, the research status of stripe noise simulation can be applied through various approaches, generally including physics-based modeling, statistical modeling, and deep learning-based methods.

1) Physics-Based Stripe Noise Simulation: Most of the current remote sensing imaging technologies rely heavily on line-scan push-broom imaging sensors [27]. Line-scan sensors encompass an array of interconnected charge-coupled devices (CCDs) in which CCD elements are arranged in a direction perpendicular to the flight path for sensing ground radiative responses and their conversion into voltage signals [28]. Subsequently, the line-scan push-broom imaging system converts these voltage signals into digital signals to generate remote sensing images; however, due to the simultaneous conversion of photoelectric signals by multiple elements, variations in sensitivity among these elements often give rise to stripe noise [29].

Specifically, the reasons for stripe noise generation stem from two main factors. On the one hand, it is due to discrepancies in the response functions of individual CCD elements, attributed to factors such as internal component system errors, sensor manufacturing process variations, and

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aging of spectrometer components. On the other hand, spatial electromagnetic interference, sensor motion, and disparities in forward and reverse scan responses all affect the illumination of the imaging system, subsequently influencing the operational state of the line-scan detector, leading to inconsistent responses among CCD elements and the emergence of stripe noise [8], [30].

In light of these challenges, it is possible to simulate the response errors on CCD elements and generate stripe noise images by constructing an effective physical model [31], [32]; however, the presence of numerous instrument-related or natural interferences during the signal conversion process makes it difficult to achieve accurate physical modeling and the characterization of stripe noise properties. Consequently, the precise simulation of stripe noise through physical modeling remains challenging in practical applications.

2) Statistical-Based Stripe Noise Simulation: Compared with physical-based methods that are complex and difficult to model, the use of statistical methods for simulating stripe noise has been widely recognized as a conventional approach. These methods simulate stripe noise by analyzing the statistical characteristics of stripe noise in actual remote sensing images and establishing corresponding statistical models [33]. For instance, Liu et al. [8] and Kuang et al. [34] have argued that, in spite of variations in the distribution of stripe noise among different sensors, they share common structural features. Specifically, degraded pixels along the direction of the stripe exhibit similar noise intensities. Consequently, they employed zero-mean Gaussian white noise added column-wise to generate stripe noise images. He et al. [35] used a polynomial model to simulate stripe noise and determine its strength based on the highest order coefficient of the polynomial. While these methods have made some significant advancements in simulating stripe noise [36] using information such as grayscale statistics, frequency spectra, and spatial correlations of the images, none of them have been able to provide the positional information of each stripe noise in the images. Additionally, due to variations in images acquired from multispectral sensors, there is still considerable room for improvement in the generality of statistically based simulation methods.

3) Deep Learning-Based Stripe Noise Simulation: By training deep neural networks, the distribution and features of stripe noise in images can be learned, allowing the generation of synthetic images with similar stripe noise. Reverse operation using deep learning-based stripe noise removal methods [37], [38], [39] can produce realistic stripe noise and offer flexibility and generalization capabilities. Compared to Statistical-based methods that rely on a large amount of prior knowledge to define features, these methods have significantly improved simulation accuracy and quality; however, they still have not focused on the issue of controlled positioning of stripe noise in images.

Overall, current research primarily focuses on the accuracy and diversity of stripe noise simulation. Researchers strive to improve the statistical parameters of models, enhance the realism of simulated results, and simulate different types and intensities of stripe noise. Additionally, some studies explore deep learning-based methods using generative adversarial networks (GANs), variational auto-encoders, and other models to achieve more accurate and flexible stripe noise simulation; however, further research is still needed to address the issue of controlled positioning of stripe noise.

B. Stripe Noise Detection

Stripe noise detection is an important research area in remote sensing image processing as it directly affects data quality and the accuracy of interpretation, ensuring the reliability of remote sensing data in various applications. Currently, research on stripe noise detection is primarily categorized into the following aspects.

1) Empirical Methods: Some empirical methods detect stripe noise by analyzing image statistical features. For example, Pei et al. [9] have used features such as gray-level co-occurrence matrix and histogram to describe the statistical characteristics of stripe noise and the application of thresholds or rules for detection. Liu et al. [40] proposed an adaptive stripe noise detection method by setting different-sized thresholds based on prior knowledge to capture the morphological characteristics of stripe noise and achieve its detection. Wang et al. [41] used calculations from multiple directions and thresholded settings to pinpoint the location of stripe noise. These methods are relatively simple and have a lot of empirical knowledge but may not perform well in complex stripe noise scenarios.

2) Frequency Domain Methods: Frequency domain methods use the spectral information of images to detect stripe noise. For instance, Zhang et al. [42] employed the ratio of row variance and column variance in the spectral image to pinpoint the location of stripe noise; furthermore, some other studies [6], [14], [43] transformed images to the frequency domain using fast Fourier transform (FFT) and analyzing the changes in the spectrum to detect stripe noise. For some periodic and obvious stripe noise, they exhibit distinctive oscillation patterns at specific frequencies in the frequency domain. Consequently, frequency domain methods can effectively detect these periodically evident stripe noise; however, there are limitations to frequency domain methods when dealing with cases where noise is mixed with image content.

3) Model-Based Methods: Some studies adopt statistical models or machine learning methods to detect stripe noise [13], [44]. For example, using auto-regressive models, hidden Markov models, or support vector machine (SVM) to model and classify stripe noise. These methods can more accurately identify specific complex stripe noise patterns. Issues such as the lack of robustness in detecting multiple-source data with manually defined features and high computational complexity, however, need to be addressed for further improvements in detection efficiency and accuracy.

4) Deep Learning-Based Methods: In recent years, deep learning methods have made significant progress in stripe noise detection [17], [45]. By using deep learning models such as convolutional neural networks (CNNs) or GANs, it is possible to learn image feature representations and noise distributions, leading to accurate stripe noise detection and

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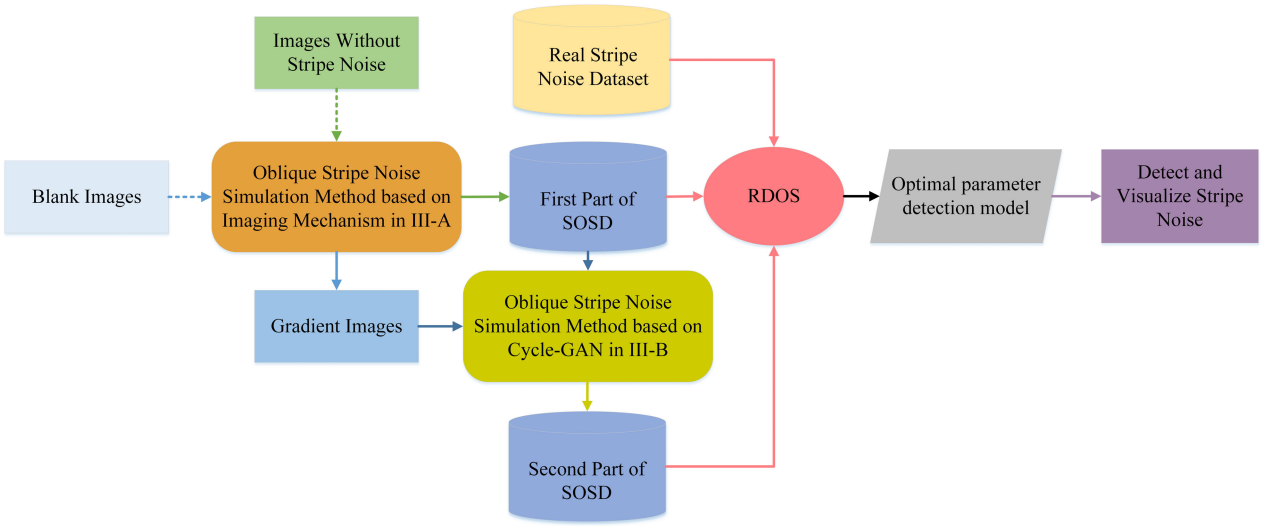


Fig. 2. Flowchart of oblique stripe noise detection. The dashed arrow signifies that only the data originating from the arrow's starting point is subjected to the specific method indicated by the arrow.

restoration. In comparison to the above methods, deep learning approaches present distinct advantages. They empower models to autonomously extract feature information from images, devoid of the constraints of manually defined limitations. Deep learning methods have, furthermore, achieved excellent results with the support of large-scale datasets and powerful computational resources.

In summary, research on stripe noise detection has made some progress; however, there are still challenges to address, such as identifying complex noise patterns similar to background features, acquiring diverse datasets to improve model generalization capabilities, and detecting oblique stripe noise that appears after image correction.

III. Methods

Research is conducted on the detection of oblique stripe noise in high-resolution calibrated remote sensing images. This research involves several key steps: First, based on the imaging mechanism of stripe noise and the theory of GANs, two methods for simulating stripe noise are designed. Second, a novel detection algorithm called "RDOS" is proposed, taking into consideration the presence of oblique stripe noise. Subsequently, a large dataset of simulated oblique stripe noise (SOSD) is generated using the two simulation methods. The RDOS algorithm is trained and evaluated on both the simulated and real stripe noise image datasets. Finally, a detection model that considers oblique stripe noise is obtained, as shown in Fig. 2 of the workflow diagram.

A. Oblique Stripe Noise Simulation Method Based on Imaging Mechanism

Oblique stripe noise, in comparison to vertical stripe noise, exhibits variations in length, angle, and randomness of the central point; therefore, simulating this type of stripe noise requires considering the diversity of central positions, as well as the variability in length and the fundamental characteristics of stripe noise. Based on these characteristics and considerations, the method for simulating stripe noise in an $M \times M$ image is as follows.

1) Randomly select $\mathbf{K}$ breakpoint coordinates on the central line in the row direction of the image, with each point represented as $\mathbf{K}_i$. The value of $\mathbf{K}$ is chosen from 1 to 5 in this study.
2) Generate $\mathbf{K}$ sets of random numbers $[\alpha_i, \beta_i, \theta'_i]$ (the subscript 'i' represents the index) within the ranges [0.75,1.25], [−5,5], [−1,1] and compute the arctangent of all θ'_i values to obtain the corresponding θ_i values (To prevent the occurrence of simulated stripe angles in decimals, we have implemented rounding in the code. Please refer to Algorithm 1, lines 3–4)
3) For each breakpoint $\mathbf{K}_i$, calculate the per-row offset value along the image's column direction, and the offset value for each row is denoted as $offsetX_j$, where the subscript "j" represents the row index. Based on the offset value for each row, obtain the corresponding offset point coordinates in the row and column directions separately. ($offsetX_j = ((M/2)\text{-}\ Row_j) \times \tan\theta_i$, where M represents the image size, Row_j represents the current row being traversed, and θ_i represents the random number corresponding to each breakpoint. During the calculation of $offsetX_j$, if ($\mathbf{K}_i + offsetX_j$) < 0, the horizontal coordinate of the offset point is 0; if ($\mathbf{K}_i + offsetX_j$) > M, the offset point is deleted). Then, for each breakpoint $\mathbf{K}_i$, connect the offset point coordinates to form a straight line with the breakpoint $\mathbf{K}_i$ as the center points, and finally generate $\mathbf{K}$ straight lines.
4) Based on the left and right vertical boundary of the image and the $\mathbf{K}$ connected straight lines in 3), divide the image into x +1 blocks along the image's row direction, and each block in denoted as C_i, where i represents the block number. Starting from the second block, for each block C_{i+1}, multiply all pixel values within the block by the α_i in 1) and add with β_i in 1), until all the blocks are calculated.

The pseudocode of the method for simulating oblique stripe noise based on the imaging mechanism is shown in Algorithm 1.

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Algorithm 1 Simulation Method of Oblique Stripe Noise Based on Imaging Mechanism

Input: *Array* images without stripe noise, *n* number of stripe noise, *Size* Image size
Output: An image with the specified number of stripe noise

```
function GENERATESIMULATENOISEFUCTION(Array, n, Size)
result ← 0, lastscale ← 1.0
offsetX_R ← randint((−size/2), (size/2) )
RotateSigma ← arctan (offsetX_R × (2/size))
for i in range (1, n + 1) do
  if i ≡ 0 then
    l ← i ×⌊(rand() × size)⌋
    r ← (i + 1) × ⌊(rand() × size)⌋
    if r ≡ 0 then r ← r + 1
    end if
  elseif i ≠ n − 1 then
      l ← r
      r ← ⌈(rand() × (Size − r)) + l⌉
  else
      l ← r
      r ← Size
  end if
  if l ≡ Size then break
  end if
  Off scale ← (rand() − 0.5) × 0.5 + 1
  OffsetValue ← randint(−5, 5)
  if |Offscale−lastscale| < 0.2 then
      off scale ← lastscale + 0.2
  end if
  lastscale = off scale
  xmax = ⌊l + (size/2) × |tanh(RotateSigma)|⌋
  xmin =⌊l -(size/2) × |tanh(RotateSigma) |⌋
  forrows in range(Size) do
      OffsetX ← ( (size/2) − rows) × tanh(RotateSigma)
      xl ← ⌊(l + OffsetX )⌋
      xr ← ⌊(l + OffsetX)⌋
      if xl < 0 then xl ←0
      end if
      if xl > Size then continue
      end if
      if xr < 0 then continue
      end if
      if xr > Size then xr ←0
      end if
      Array[rows, xl: xr, :] ← Array[rows, xl: xr, :] × Off scale + OffsetValue
    end for
  end for
 return Array
end function
```

The above simulation method for oblique stripe noise can only generate oblique stripe noise within the range $(-(\Pi/4), (\Pi/4))$; therefore, after generating the image, rotate it 90° clockwise with a probability of 0.5. This allows for the simulation of all possible oblique stripe noise data within the range $(-(\Pi/2), (\Pi/2))$.

B. Oblique Stripe Noise Simulation Method Based on Cycle-GAN

In Section III-A, an introduction was provided on simulating stripe noise based on the imaging mechanism of oblique stripe noise. Although that method has the capability to produce a multitude of oblique stripe noise images with diverse orientations and shapes, it is limited by the existing background features in remote sensing data. Consequently, numerous simulated oblique stripe noise images generated still exhibit similar background elements, and there is still substantial potential for enhancing the diversity of background elements in simulating the effects of oblique stripe noise.

In recent years, with the development of deep learning and the introduction of GANs, their applications in the field of image generation have been extensive [46]. These applications include high-resolution image generation, handwritten Chinese character generation, video prediction, and facial image generation, among others [47], [48], [49]. Using GANs for image generation, therefore, serves the purpose of data augmentation and is highly achievable. At the same time, in the field of remote sensing, numerous researchers have harnessed and improved upon GAN-based techniques [50], [51], achieving significant progress in various areas of remote sensing generation, such as image super-resolution reconstruction and map transformation. This further underscores the potential of GAN-based approaches as an excellent choice for addressing remote sensing generation challenges.

In the context of oblique stripe noise generation, our aim is to generate diverse oblique stripe noise images of background terrains based on controllable stripe noise information and serve to enhance the augmentation of our dataset. Given these considerations, we have selected the Cycle-GAN algorithm to train an oblique stripe noise image simulation model [21]. Cycle-GAN, a generative network architecture widely embraced by researchers across diverse fields, has undergone significant refinements, leading to substantial advancements [52], [53], [54], [55], [56]. This extensive adoption and successful evolution establish it as a reliable algorithm; furthermore, the algorithm's functional characteristics are noteworthy. It excels in training neural networks for image-style transformations without necessitating paired data. Additionally, it incorporates cycle consistency constraints to prevent the arbitrary generation of output images. These features effectively address the challenges posed by the aforementioned oblique stripe noise in the generation process. Consequently, the specific procedures of our study are outlined as follows.

First, creating a lot of blank images with RGB values all set to 128. Then, the mechanism described in Section III-A of simulating oblique stripe noise based on the imaging mechanism is applied to convert the blank images into gradient images. Next, the gradient images and the high-resolution remote sensing images with oblique stripe noise based on the imaging mechanism from Section III-A are used as inputs to the Cycle-GAN algorithm, and optimization is performed based on the cycle-consistency loss. Finally, the trained generator is used to generate a large number of new images with oblique stripe noise. The overall process is illustrated in Fig. 3.

C. RDOS Algorithm

The RDOS algorithm is an improvement upon the LSND algorithm [17]. This improvement is primarily to address the increased complexity in length and angle of oblique stripe noise detection compared to the horizontal/vertical stripe noise in the original images after image correction, which is difficult to apply based on the LSND algorithm.

Specifically, in terms of the RDOS network framework, we have used the YOLOv5 [57] network model's backbone and made improvements to the output end of the LSND algorithm. For the output end part, we have employed k-means clustering [58] to obtain multiple prior lengths suitable for

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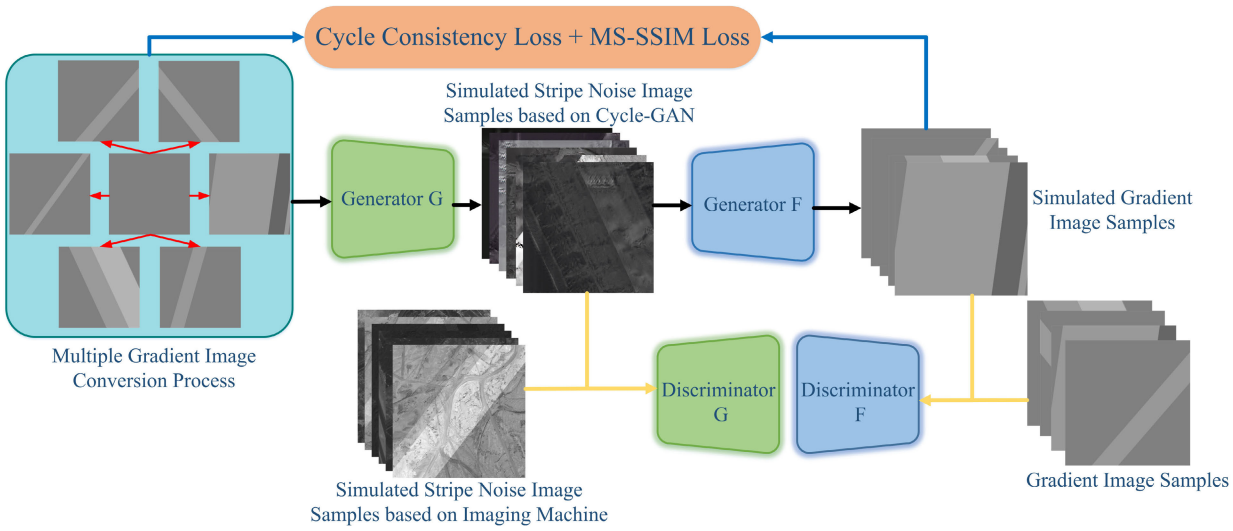


Fig. 3. Flowchart of simulated oblique stripe noise training based on Cycle-GAN.

stripe noise. Then, we have modified the network structure of the output part so that the RDOS-based model has the function of predicting the stripe noise angle.

Additionally, we have employed circular shifted labeling (CSL) encoding [59] to represent the rotational degree in the sample labels. The CSL approach treats the angle as a classification problem, simplifying model convergence; moreover, considering that the precision requirements for line-based object detection are much higher than general object detection, the CSL approach yields more prominent results compared to using one-hot encoding for the sample labels and achieves a more balanced distribution between positive and negative samples, resulting in a more robust training outcome.

In the loss function part, we introduced angle-related losses in the loss functions and increased the number of predict lines within each predict grid cell. Last, the postprocessing stage of the RDOS algorithm, such as linear nonmaximum suppression, is consistent with the LSND algorithm.

1) Network Structure Framework: The RDOS network framework improves upon the output end of the LSND algorithm with the following specific modifications.

First, the lengths of all stripe noise samples are collected, followed by applying the k-means algorithm to obtain three cluster centers representing the three prior lengths for stripe noise. Then, compared with LSND's output part, in addition to the four basic parameters (x, y, h, confidence) and the predicted stripe class for each detected line, the framework includes an additional component for predicting the rotation angle. The angle prediction is treated as a classification task, employing 180 channels to represent the discerned tilt angles of oblique stripe noise. This results in an output tensor of dimensions $3 \times (4 + 1 + 180) = 555$, as shown in Fig. 4.

2) Loss Function: Regarding the rotation angle feature, the sample labels are encoded using CSL encoding, and the angle loss is calculated using the cross-entropy loss function. The remaining parts remain unchanged from the LSND algorithm. The total loss function is as follows:

$$
\begin{aligned}
&\text{LOSS} \\
&= \sum_{i=0}^{s^2} \sum_{\mathrm{j}=0}^{L} I_i^{\mathrm{obj}} [a_1 \left(Lx_i - \widehat{L}x_i\right)^2 \\
&+ a_2 \left(Ly_i - \widehat{L}y_i\right)^2 + \frac{a_3 \left(Lh_i - \widehat{L}h_i\right)^2}{k}] \\
&- \sum_{i=0}^{s^2} \sum_{\mathrm{j}=0}^{L} I_i^{\mathrm{obj}} \left[\widehat{C}_{\mathrm{i}} \log(C_i) + \left(1 - \widehat{C}_{\mathrm{i}}\right) \log(1 - C_i)\right] \\
&- \lambda_{\mathrm{noobj}} \sum_{i=0}^{s^2} \sum_{\mathrm{j}=0}^{L} I_i^{\mathrm{noobj}} \left[\widehat{C}_{\mathrm{i}} \log(C_i) + \left(1 - \widehat{C}_{\mathrm{i}}\right) \log(1 - C_i)\right] \\
&- \sum_{i=0}^{s^2} \sum_{\mathrm{j}=0}^{L} I_i^{\mathrm{obj}} \sum_{c \in \mathrm{classes}} \left[\widehat{P}_i \log(p_i) + \left(1 - \widehat{p}_{\mathrm{i}}\right) \log(1 - p_i)\right] \\
&- \sum_{i=0}^{s^2} \sum_{\mathrm{j}=0}^{L} I_i^{\mathrm{obj}} \sum_{k \in \mathrm{angles}} \left[\widehat{R}_i \log(R_i) + \left(1 - \widehat{R}_{\mathrm{i}}\right) \log(1 - R_i)\right].
\end{aligned}
\tag{1}
$$

The RDOS algorithm's loss function comprises four components: localization offset loss, confidence loss, angle loss, and classification loss. The key implementation of the loss function is its emphasis on dividing the feature map of the image for detection into an $S \times S$ grids, ensuring efficient and comprehensive object detection across the grid cells.

Specifically, localization offset loss, the first component, is computed using the mean squared error and evaluates how well the predictions align with the actual positions and size of stripe noise. In this context, I_i^{obj} indicates whether the ith grid cell contains the center point of stripe noise (if it is true, it equals 1; otherwise, it equals 0); S^2 represents the number of grid cells the image is divided into, while j represents the number of predict lines for each grid cell; Lx_i, Ly_i, represent the predicted center coordinates of stripe noise in the ith grid cell along the row and column directions,

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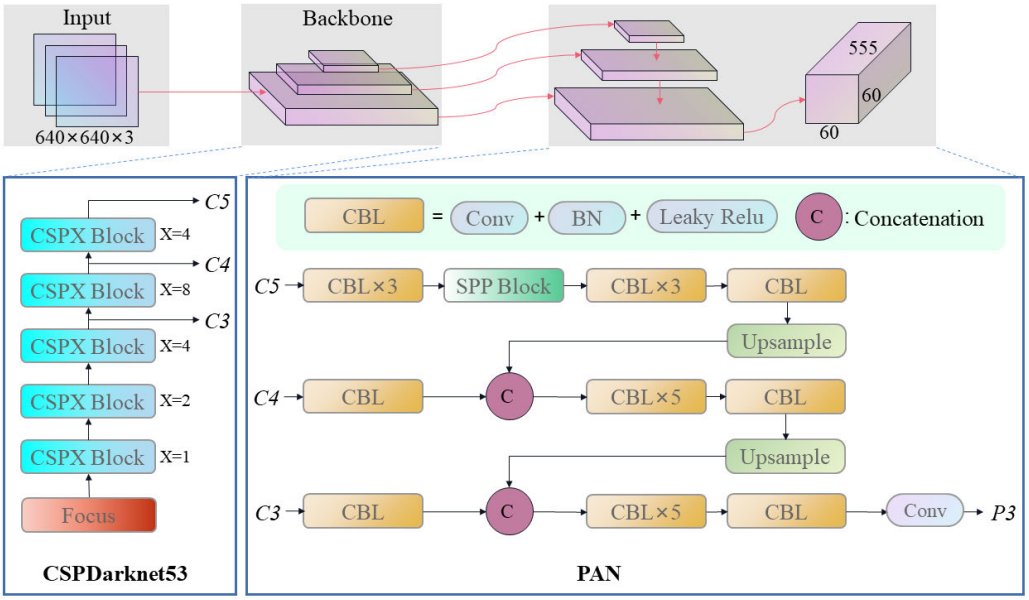


Fig. 4. RDOS algorithm network model framework.

while Lh_i represents the length of the predicted stripe noise; $\widehat{Lx}_i$, $\widehat{Ly}_i$, and $\widehat{Lh}_i$ represent the actual center coordinates and length of stripe noise. Additionally, α_1, α_2, α_3, and k are used to balance scales. The second and third components pertain to confidence loss, which employs two cross-entropy loss functions. This part assesses how well the confidence predictions align with the actual presence or absence of stripe noise. In this context, I_i^{noobj} indicates whether the ith grid cell lacks the center point of stripe noise; C_i represents the confidence score, and $\widehat{C}_i$ signifies the label for the presence of stripe noise; λ_{noobj} is added to reduce the contribution of the loss value for grids without objects to the total loss. The fourth component and the fifth component are classification loss and angle loss, which all also use cross-entropy loss function to determine how well predictions match the actual object categories and the degree of mismatch between the angle information of stripe noise within each grid cell and the actual angles. Among them, C represents the category; P_i represents the confidence score for belonging to category C, and $\widehat{P}_i$ denotes the actual category label; K represents the degree of angles (integer values in the range of 0–180); R_i represents the confidence score for the angle K, and $\widehat{R}_i$ represents the actual angle label for stripe noise.

IV. Experiment and Analysis

To simulate a large number of oblique stripe noise images using the two methods described in Sections III-A and III-B and train and evaluate the RDOS algorithm, the following steps were taken on an Ubuntu 18.04 LTS operating system with an NVIDIA Quadro M4000 GPU with 16 GB of VRAM. The software packages used include Visual Studio Code, Python 3.7.4, PyTorch 1.7.1, CUDA 10.1, and cuDNN 7.3.1.

A. Experimental Dataset and Model Training

This study selected nine scenes of GaoFen-1 (GF-1) and GF-2 images, which were without oblique stripe noise, and 1 scene of images with oblique stripe noise. First, due to the large size of TIFF-format image data, the images were converted to JPG format to reduce the data size. Then, the transformed images were divided both horizontally and vertically, with 640-pixel intervals, resulting in multiple basic sample images sized at 640 × 640 pixels. Among them, there were 406 images containing real stripe noise. Finally, a large number of simulated oblique stripe noise data was generated using the two aforementioned simulation methods based on the experimental data without oblique stripe noise.

For the Cycle-GAN simulated oblique stripe noise images, due to hardware limitations, this study scaled the input images to 320 × 320. The number of gradient images (TrainA) is 4305, and the number of simulated oblique stripe noise images based on the imaging mechanism (TrainB) is 5547. The training and validation split ratio was 9:1, and the training was conducted for 15 epochs, with the remaining configuration parameters consistent with the original Cycle-GAN paper. Finally, the trained generator was used to generate images, and the generated images were interpolated to 640 × 640 using the bilinear interpolation method.

When constructing an experimental dataset, it is necessary to establish the ratio for selecting two types of simulated stripe noise data; we aim to strike a balance between the following two crucial factors. On the one hand, our aim is to introduce a greater degree of variability rooted in actual sample variations within the dataset, ensuring that the training data aligns with real-world scenarios; therefore, the data part obtained based on the image mechanism needs to be the main part of the simulated data.

On the other hand, we consider the diversity of background features in the simulated data to cover a wider range of scenarios in the entire dataset. So, we need to ensure that the simulated data based on the Cycle-GAN-based model occupies a certain amount in the dataset. As a result, we opted for an 8:2 ratio to combine these two forms of simulated stripe noise data, integrating them with real stripe noise data to

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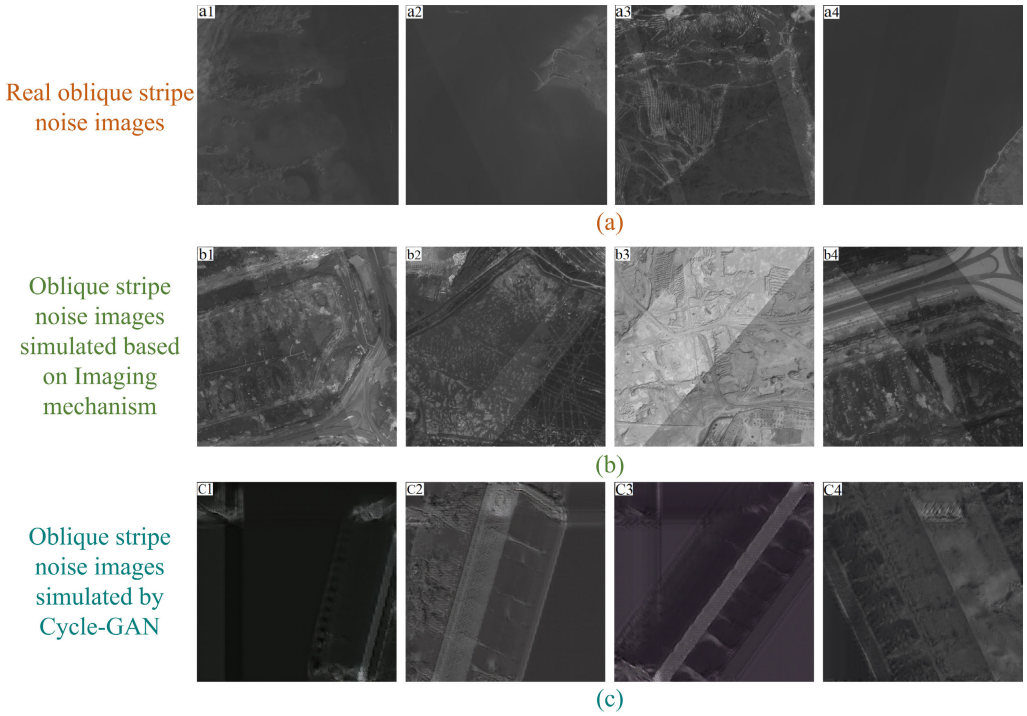


Fig. 5. Comparison of simulated oblique stripe noise images and real images. (a) Real oblique stripe noise images. (b) Simulated oblique stripe noise images based on the imaging mechanism. (c) Oblique stripe noise images simulated by Cycle-GAN. Each category's subfigures represent schematic schematics of oblique stripe noise with various background elements, lengths, and angles.

TABLE I

LOCATION INFORMATION OF STRIPE NOISE. EACH ROW REPRESENTS THE CATEGORY, CENTER POINT COORDINATES, LENGTH, AND ANGLE OF THE STRIPE NOISE IN THE IMAGE (ANGLE VALUE IS BASED ON THE DIRECTION OF THE IMAGE ROW AND IS CALCULATED CLOCKWISE)

Category	Center point coordinates X	Center point coordinates Y	Length H	Angle
0	514.58	320	641.01	113
0	567.58	320	370.67	113
0	570.58	320	355.31	113

construct the experimental dataset. The final composition of the experimental dataset is illustrated in Fig. 5.

From Fig. 5., it can be observed that both methods effectively retain the characteristics of the stripe noise. Especially in the samples generated by the Cycle-GAN as the baseline method, compared to the method based on remote sensing principles, there is a significant addition of background features, which enhances the diversity of the samples.

During the simulation process, the coordinates of the center point, length information, and angle information of the stripe noise in each image were saved and then converted into corresponding labels, as shown in Table I.

Finally, all the simulated stripe noise images and labels are divided into a training set and a validation set with a ratio of 19:1. The training set consists of **149 672** images, and the validation set consists of **7875** images. During the actual training process, various data augmentation techniques such as random cropping, flipping, and color space transformations are applied to the images in the training set to further enhance sample diversity during model training.

B. Loss Function Curve Analysis

Based on the oblique stripe noise dataset described in Section IV-A, this study trained the RDOS model proposed in Section III-C. The exponential moving average (EMA) method was used to average the model parameters to improve the model's test metrics and robustness. The batch size was set to 4, the number of training epochs was 50, and the initial learning rate was set to 0.001. The learning rate was reduced using the cosine annealing technique [60]. The Adam optimizer [61] was used with the parameters β_1 initialized to 0.937, β_2 initialized to 0.999, and weight decay set to 0.005. The model parameters were updated every 16 batches. During the first three epochs, a warm-up [62] was applied to the learning rate and the β_1 parameter of the Adam optimizer. This means that during the first three epochs, the learning rate linearly decreased from 0.1 to the learning rate obtained using the cosine annealing strategy, and the β_1 parameter linearly increased from 0.9 to 0.937. For the remaining training epochs, the learning rate decayed according to the cosine annealing strategy, while the β_1 parameter remained constant at 0.937. The training loss curves for the localization offset loss, confidence loss, angle loss, and total loss in the validation set were plotted and shown in Fig. 6.

From the figure, we can observe that all losses rapidly decrease and converge within the first 4–5 epochs (indicated by blue dashed boxes). This indicates that the model is capable of achieving fast convergence during training, demonstrating its practicality. Additionally, by assigning different weight coefficients to the three loss terms, the losses are kept within the same magnitude, thereby avoiding the issue of imbalanced gradients affecting different types of losses.

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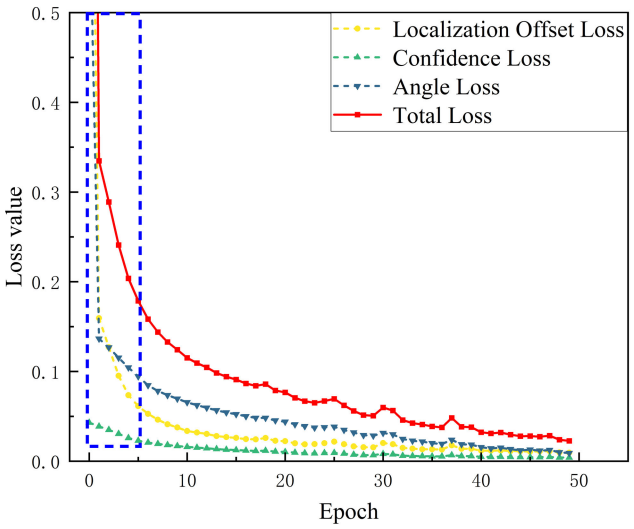


Fig. 6. Loss curve chart based on RDOS training. Among these, the four curves in yellow, green, blue, and red, respectively, represent localization offset loss, confidence loss, angle loss, and total loss during the training process.

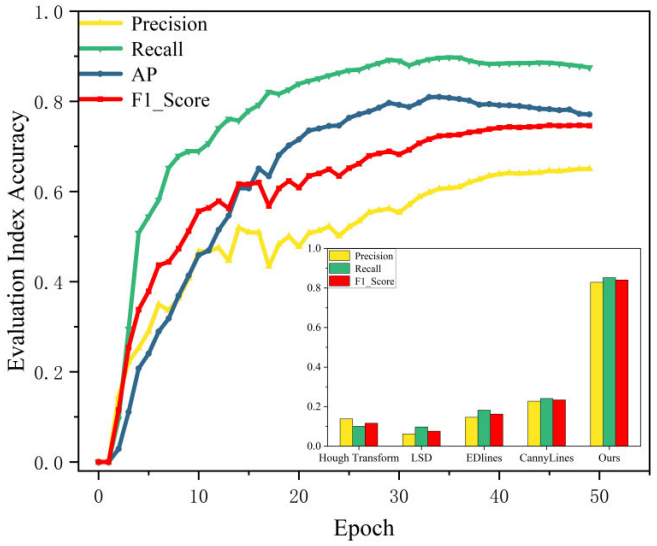


Fig. 7. Accuracy curves of various indicators. The histogram in the lower right corner shows the comparison between the classic line detection models and our model on various detection indicators.

C. Accuracy Evaluation Analysis

For the line-shaped object detection problem, this study selected common model evaluation metrics and methods for choosing confidence threshold for linear object prediction. These were applied to evaluate and compare the models.

1) Accuracy Evaluation Index and Prediction Confidence Threshold Selection Method: Precision, recall, average precision (AP), and $F1$ score are basic concepts and metrics in object detection. The calculations for these parameters are shown in the following equations:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{2}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3}$$

$$\text{AP} = \frac{1}{C} \sum_{K=i}^{N} P(k) \Delta r(k) \tag{4}$$

$$F1\text{Score} = \frac{2\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5}$$

where TP represents the number of correctly identified stripe noise, FP is the number of backgrounds falsely identified as stripe noises, FN represents the number of stripe noises not identified, N represents the total number of stripe noises to be detected, $P(k)$ represents the precision value when k stripe noises are identified, and $\Delta r(k)$ represents the change in recall value when the number of identified stripe noises changes from $k-1$ to k (by adjusting the threshold).

This study, moreover, focuses on linear object detection, so IoU, DIoU, and CIoU [63], etc. cannot be used as metrics for evaluating the prediction confidence threshold; therefore, in terms of threshold setting, two considerations need to be taken into account. On the one hand, linear objects differ from traditional 2-D image objects as they only possess 1-D characteristics. On the other hand, while a pixel error in detecting objects within a bounding box is considered negligible, in the case of predicting linear objects, such an error is considered a detection failure. Hence, it is important to ensure the accuracy of the judgment between predicted and actual results [17]. As a result, the threshold is set as follows.

For an $M \times M$ image, the threshold is set as $(M - 1/M)$. When the ratio of the area surrounded by the predicted line and the actual line to the total image area is less than $(1/M)$, the stripe noise is considered correctly identified; otherwise, it is considered a misidentification. Additionally, the interval $[(M - 1/M), (10 \times (M-1)/10 \times M)]$ is used as the threshold for determining samples in the calculation of AP.

2) Accuracy Evaluation and Comparative Analysis: For model evaluation metrics and prediction confidence threshold, this study adopts the metrics of (2)–(5). The accuracy, recall, $F1$-score, and AP curve on the validation set are shown in Fig. 7.

From the figure, it can be observed that the evaluation metric curves obtained using RDOS converge rapidly. Once converged, the metrics show minimal fluctuations, indicating good performance across all evaluation metrics.

Additionally, this study also selected several mainstream straight-line detection methods as comparative experiments for the detection of stripe noise, including the Hough Transform [64], LSD algorithm [65], EDlines algorithm [66], and CannyLines algorithm [67]. These conventional line detection methods heavily rely on manual feature design and selection, which require extensive experience and time for feature extraction. These methods, therefore, lack strong generality and are constrained in terms of objectivity, robustness, and generalization. The mechanism behind stripe noise generation and the approach of straight line detection are, moreover, significantly different, leading to substantial differences in the characteristic forms and numerical distributions of strip noise in images. As a result, in Table II, the detection accuracy of these methods, as indicated by various metrics, is generally below 30% and is challenging to apply in practical engineering

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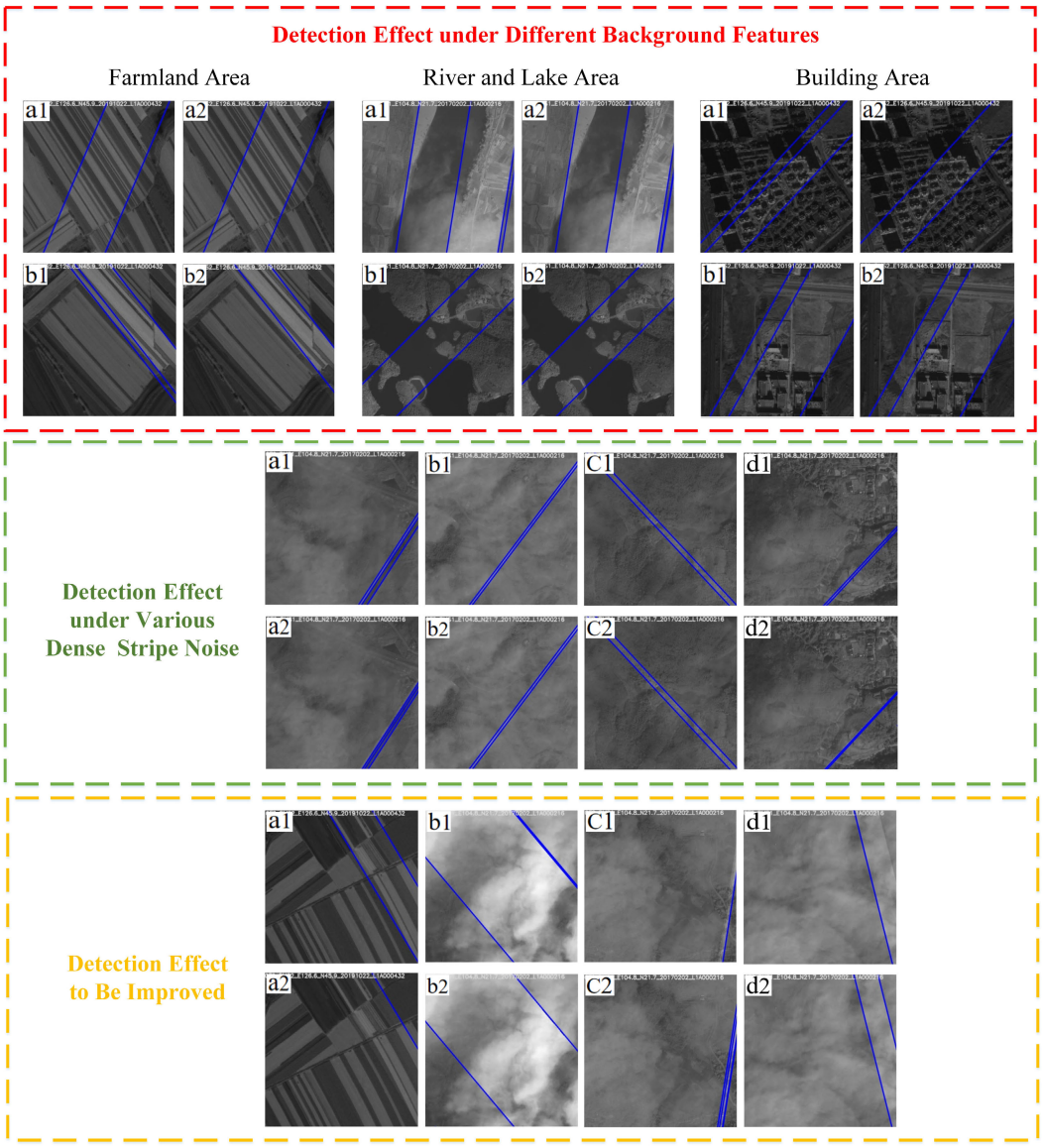


Fig. 8. Comparison of detection effect between real and predicted results under different circumstances. The red dotted box shows the detection results under different background features (farmland area, river and lake area, and building area); the green box showcases results under various dense stripe noise; and the yellow box highlights areas where further improvement is needed for detection results. Specifically, paired images labeled '1' and '2' beneath each English letter (a)–(d) depict the contrast between real and predicted results.

scenarios. In contrast, the RDOS-based model can effectively detect various forms of stripe noise in complex backgrounds of remote sensing images, showcasing its accuracy and robustness.

Finally, this study adjusted the confidence threshold for the optimized parameter model obtained from the RDOS algorithm. The precision achieved was 82.93%, the recall rate was 85.17%, the $F1$ score was 84.04%, and the AP value was 82.34%.

D. Accuracy Comparison of Models Under Different Data Sizes

To further validate the impact of the two simulated data methods on the experimental results, this study trained the RDOS algorithm on different-scale datasets and compared the accuracy using the validation set from Section IV-A. The accuracy comparison is shown in Table III.

From the various evaluation metrics, it can be observed that the RDOS-based model shows improvements in accuracy across all metrics when the simulated stripe noise datasets generated using the two simulation methods are added; furthermore, as the dataset size increases, the model becomes more effective in detecting stripe noise in complex background scenarios. This demonstrates the effectiveness of adding simulated data to enhance accuracy. Additionally, the RDOS algorithm exhibits better performance on large-scale datasets, highlighting its robustness and scalability.

E. Detection Results Discussion

To further analyze and discuss the performance of the optimal detection model based on the RDOS algorithm, this

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TABLE II
QUANTITATIVE ACCURACY COMPARISON BETWEEN THE CLASSIC LINE DETECTION MODELS AND OUR MODEL ON VARIOUS DETECTION INDICATORS. BOLD NUMBERS REPRESENT THE BEST VALUES IN EACH COLUMN

Methods	Precision (%)	Recall (%)	F1 Score (%)
Hough Transform	13.61	9.78	11.30
LSD	5.95	9.45	7.27
EDlines	14.34	17.97	15.93
CannyLines	22.53	23.92	23.19
Ours	**82.93**	**85.17**	**84.04**

TABLE III
ACCURACY COMPARISON UNDER DIFFERENT DATA SCALES. EACH ROW IN THE TABLE REPRESENTS THE OPTIMAL VALUE OF THE ALGORITHM UNDER ALL EPOCHS EVALUATED UNDER DIFFERENT DATA SCALES: GT REPRESENTS THE NUMBER OF BAND NOISE IMAGES TO BE DETECTED; REAL REPRESENTS THE REAL DATASET; SD1 REPRESENTS THE DATASET SIMULATED BY USING THE IMAGING MECHANISM; SD2 REPRESENTS THE DATASET SIMULATED BY CYCLE-GAN; BOLD NUMBERS REPRESENT THE BEST VALUES IN EACH COLUMN

Data Size	GT	Precision (%)	Recall (%)	F1 Score (%)	AP (%)
REAL	406	<50	<50	<50	<50
REAL+SD1	119,818	81.44	80.36	80.89	77.62
REAL+SD2	30,260	61.22	69.71	65.19	62.33
ALL	149,672	82.93	85.17	84.04	82.34

TABLE IV
COMPARISON OF $F1$ SCORE INDEX RESULTS BASED ON VARIOUS BACKGROUND FEATURES AND VARIOUS DENSE STRIPE NOISES. IN EACH CASE, 15 IMAGES TO BE DETECTED WERE MANUALLY RANDOMLY SELECTED

Accuracy Evaluation Index	Different background features			Dense stripe noise
	Farmland	River and Lake	Building	
F1 Score (%)	79.38	88.24	90.63	78.33

study conducted a comparative visual analysis of the detection results based on various background features and various dense stripe noise, comparing them with the ground truth labels, and performing quantitative analysis in Table IV; furthermore, subpar detection results were showcased to further explore the directions for model optimization. The visual results of the detection comparison are presented in Fig. 8.

First, the detection results and accuracy based on different background features (indicated by red dashed boxes) demonstrate that the optimal detection model using the RDOS algorithm achieves satisfactory performance for image backgrounds with linear characteristics, such as field boundaries, river boundaries, and dense building structures. This further confirms the accuracy and robustness of our model. Second, the detection results and accuracy based on various dense stripe noises (indicated by green dashed boxes) reveal our model's strong anti-interference capability in detecting adjacent stripe noise at different angles, yielding excellent detection results. Additionally, our model achieves a high detection rate of up to 33.33 frames/s under the current experimental environment, representing a 60% improvement compared to the above classical line detection models, highlighting its significant practical value in industrial and downstream applications; moreover, it offers valuable insights for addressing engineering issues related to multiangle linear object detection.

Last, this study visually compares and showcases the subpar detection results. It can be observed that the abrupt variations in the appearance of undetected stripe noise are consistent with the overall morphological distribution of the images. In such cases, the model mistakenly recognizes the characteristics of stripe noise as a special signal of the images themselves, leading to missed or false detection; moreover, analysis of the reasons for the false detection of too dense stripe noise shows that the linear nonmaximum suppression in the post-processing stage filters out the detection results. Further design and consideration are, therefore, required to optimize the model for capturing deeper-level features that differentiate from image styles and to employ more effective post-processing methods in order to enhance the algorithm's accuracy.

V. CONCLUSION

This study proposes an algorithm called "RDOS" for detecting oblique stripe noise in high-resolution remote sensing images after image correction. Specifically, the RDOS-based model construction is as follows.

First, a simulation method for generating oblique stripe noise is proposed based on the imaging mechanism of oblique stripe noise generation. A large dataset of remote sensing images with simulated oblique stripe noise, referred to as the first part of SOSD, is generated using this method.

Next, using gradient images created from blank images based on the simulation method based on the imaging machine and the first part of SOSD, we use the Cycle-GAN algorithm to train and evaluate to develop an oblique stripe noise generation model. The second part of SOSD, a new dataset of remote sensing images with oblique stripe noise, is then simulated using the generation model.

Finally, improvements are made to the classifier and loss function of the LSND network model, and a new algorithm called "RDOS," which takes oblique stripe noise into account, is proposed. The RDOS algorithm is trained and evaluated on a combined dataset consisting of the real stripe noise dataset and SOSD from the two parts.

Experimental results demonstrate that the optimal detection model based on the RDOS algorithm achieves good performance in various detection metrics and exhibits excellent speed. The algorithm shows promising detection effectiveness in practical industrial applications and provides insights for visual engineering problems such as multiangle linear object detection.

There are, however, still areas for improvement in future research. On the data aspect, on the one hand, limitation in the amount of real stripe noise data: our current study only uses a limited number of high-resolution multispectral remote sensing images provided by GF-1 and GF-2 sensors. Although we achieved some positive results, we realized that more real

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stripe noise data is needed to validate our method further. On the other hand, limitation in the number of sensor types: we have not yet conducted in-depth experiments on other types of remote sensing images, such as infrared/thermal remote sensing images and microwave-based synthetic aperture radar (SAR) images. We plan to extend this part of our research in the future to gain a more comprehensive understanding of the characteristics, differences, and commonalities of stripe noise from different sensors and to build a more comprehensive stripe noise detection technology framework and model.

We, furthermore, recognize that there is still much potential room for improvement in detecting large-scale oblique stripe noise in panoramic scenes. We hope to develop more efficient and accurate solutions to further advance the field of remote sensing in various industrial applications. This will help address stripe noise challenges in complex panoramic scenes and provide more powerful tools for professionals in related fields.

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